Diparna Santra

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Professional Summary

Statistics postgraduate with hands-on experience in Python, R, Data Analysis, Statistical Modeling, and Machine Learning. Skilled in data preprocessing, Exploratory Data Analysis (EDA) and feature engineering. Strong analytical foundation in probability, statistical inference, and simulation, with practical exposure to regression analysis and classification problems. Seeking a Data Analyst role to apply statistical and analytical skills to real-world business decision-making.

Education

Course	Institution	Year	CGPA/%
M.Sc. in Statistics	Visva-Bharati University	2024-26	7.2
B.Sc. in Statistics	Haldia Government College	2021-24	7.85
Higher Secondary	Bhangamora Nutangram K.N.C.M. Institution	2021	86.80%

Skills Summary

- **Languages**: Python, R, C, SQL
- **Libraries**: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- **Tools**: MS Excel, Powerpoint, Word, LaTeX
- **Interests**: Data Analysis, Data Visualisation, Artificial Intelligence

Master Thesis

- **Prediction of Censored Lifetimes under Proportional Hazard Rate Models — Prof. Sudhansu S. Maiti** [Link]
  - Developed statistical predictors for censored failure times using Progressive Type-II censoring and Proportional Hazard Rate (PHR) models across Exponential, Lindley, and Rayleigh distributions.
  - Built and compared multiple frequentist (MLP, BUP, CMP) and Bayesian prediction methods, validated through a 1,000-run Monte Carlo simulation in R.
  - Tested predictors on real-world ball-bearings fatigue life data, identifying the best-performing model with Rayleigh distribution yielding the highest accuracy.

Projects

- **Breast Cancer Diagnosis (Classification Analysis)— Prof. Sudhansu S. Maiti (Visva-Bharati)** [Link]
  - Processed the UCI Breast Cancer Wisconsin Diagnostic dataset (569 observations) to classify cell nucleus properties as malignant or benign for automated medical diagnosis.
  - Performed data preprocessing using Z-score **outlier detection**, 5th/95th percentile **winsorization**, **Min-Max scaling**, and pair-wise **correlation analysis** to mitigate **multicollinearity**, reducing feature dimensionality from 30 to 14.
  - Implemented and trained classification algorithms including **Logistic Regression**, **Decision Trees**, **Random Forest**, and **Support Vector Machine (SVM)** on an 80:20 train-test split.
  - Evaluated model efficacy using **Confusion Matrix**, **Precision**, **Recall**, **F1-Score**, and **ROC-AUC** metrics, selecting **SVM** as the optimal model with **97.37% accuracy** and an **AUC of 0.9957**.
- **Food Delivery Time Prediction (Regression Analysis) — Prof. Sudhansu S. Maiti (Visva-Bharati)**[Link]
  - Cleaned and preprocessed a food delivery dataset of **41,953** observations by handling missing values, extracting numerical attributes (Age, Ratings, Time taken), and cleaning categorical strings (Weather, Traffic density, City).
  - Engineered spatial and temporal features including **Haversine distance** from restaurant/delivery coordinates and order-to-pickup delay, followed by **One-Hot Encoding** and multicollinearity assessment via Variance Inflation Factor (**VIF**).
  - Implemented and trained **Multiple Linear Regression (OLS)**, **Ridge (L_2 Regularization)**, and **Lasso (L_1 Regularization)** models on an 80:20 train-test split with 5-fold cross-validation for hyperparameter (α) tuning.
  - Evaluated performance using R^2, **RMSE**, and **MAE** metrics; selected **Lasso Regression** as the optimal model for performing feature selection, identifying traffic density, distance, and weather conditions as primary drivers of delivery times.

Certificates

- *Python for Beginners* by Simplilearn SkillUp [Link]*(Code: 10383241)* — June 23, 2026
- *Introduction to Generative AI Studio* by Google Cloud & Simplilearn SkillUp [Link]*(Code: 10462953)* — July 13, 2026
- *SQL Analytics and BI* on Databricks & Simplilearn SkillUp [Link] *(Code: 10546855)* — Aug 2, 2026

Extracurricular Activities

- Actively participated and performed in various college competitions and cultural functions for **flute** performance and **recitation**.